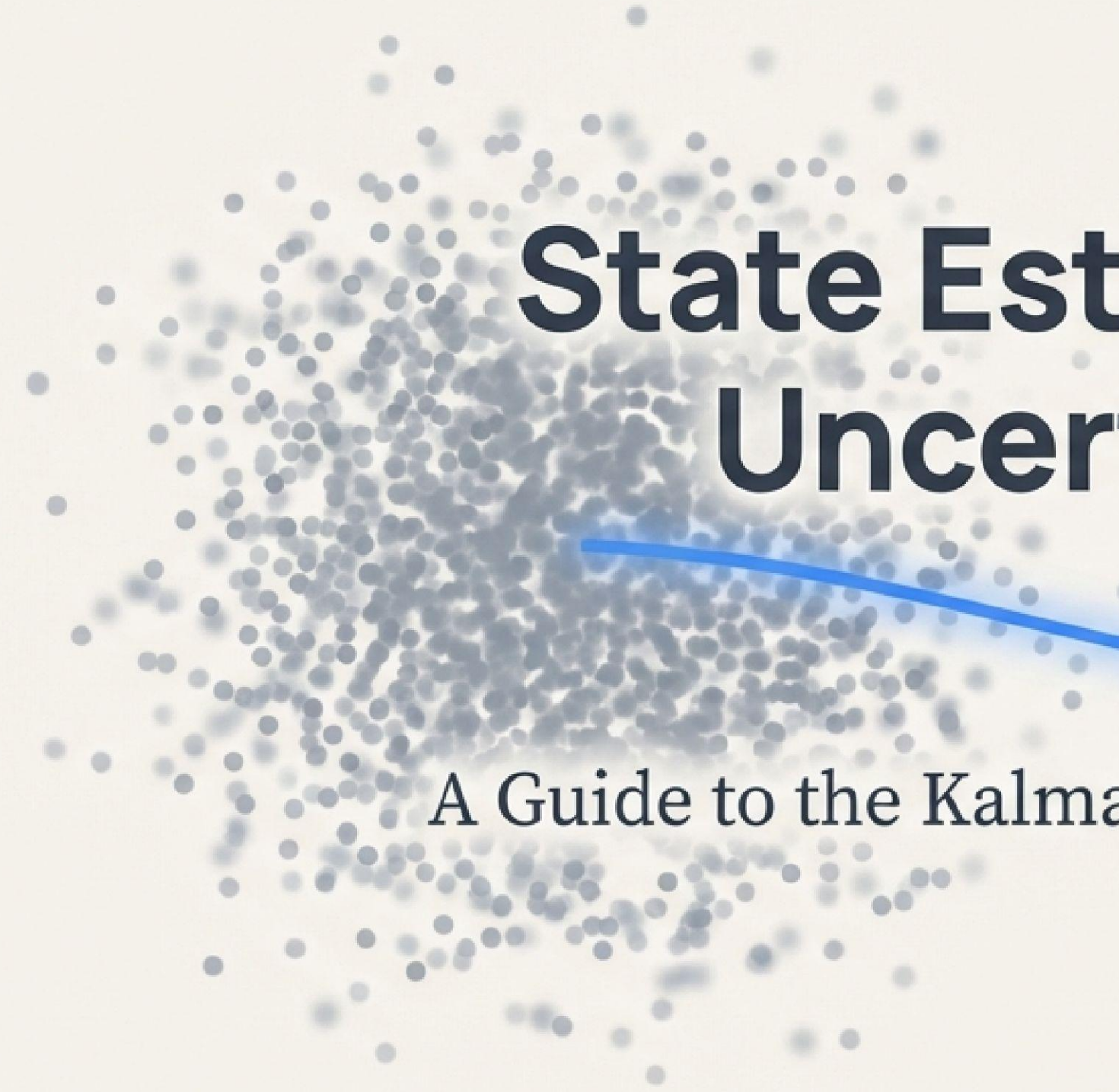




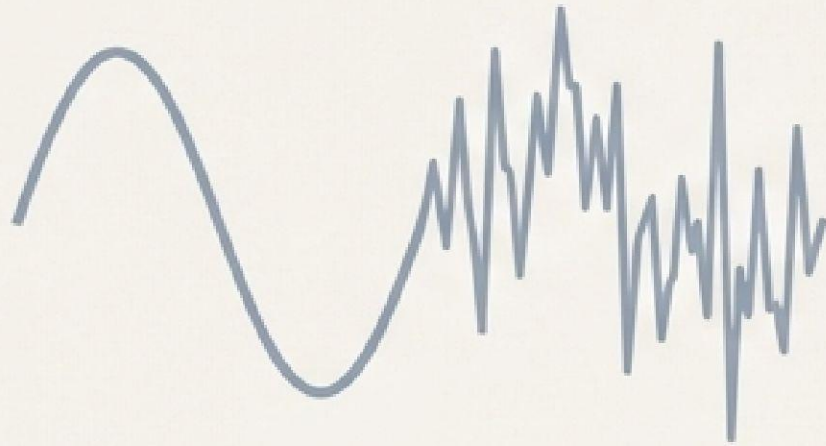
State Estimation in an Uncertain World

A Guide to the Kalman & Extended Kalman Filters



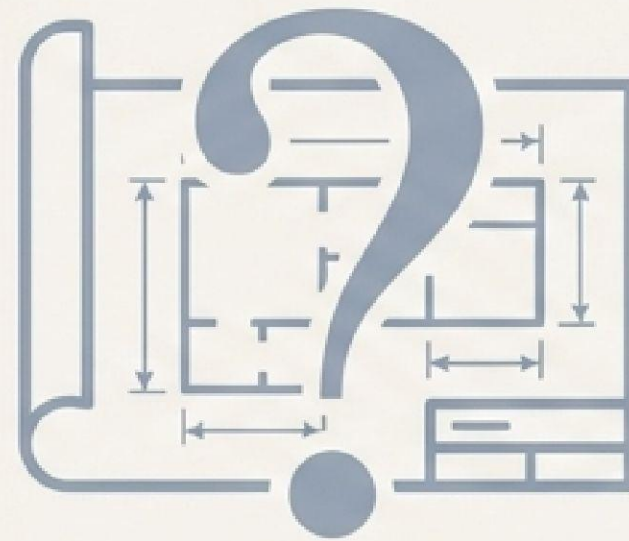
Why We Can't Simply Trust Our Data

We operate in a world where information is imperfect. The goal is to find the most accurate 'true state' of a system (like its position or velocity) when we can't observe it directly.



Noisy Sensors

Our measurements are inherently unreliable and contain errors.



Imperfect Models

The mathematical models we use to predict a system's behavior are only approximations of reality.

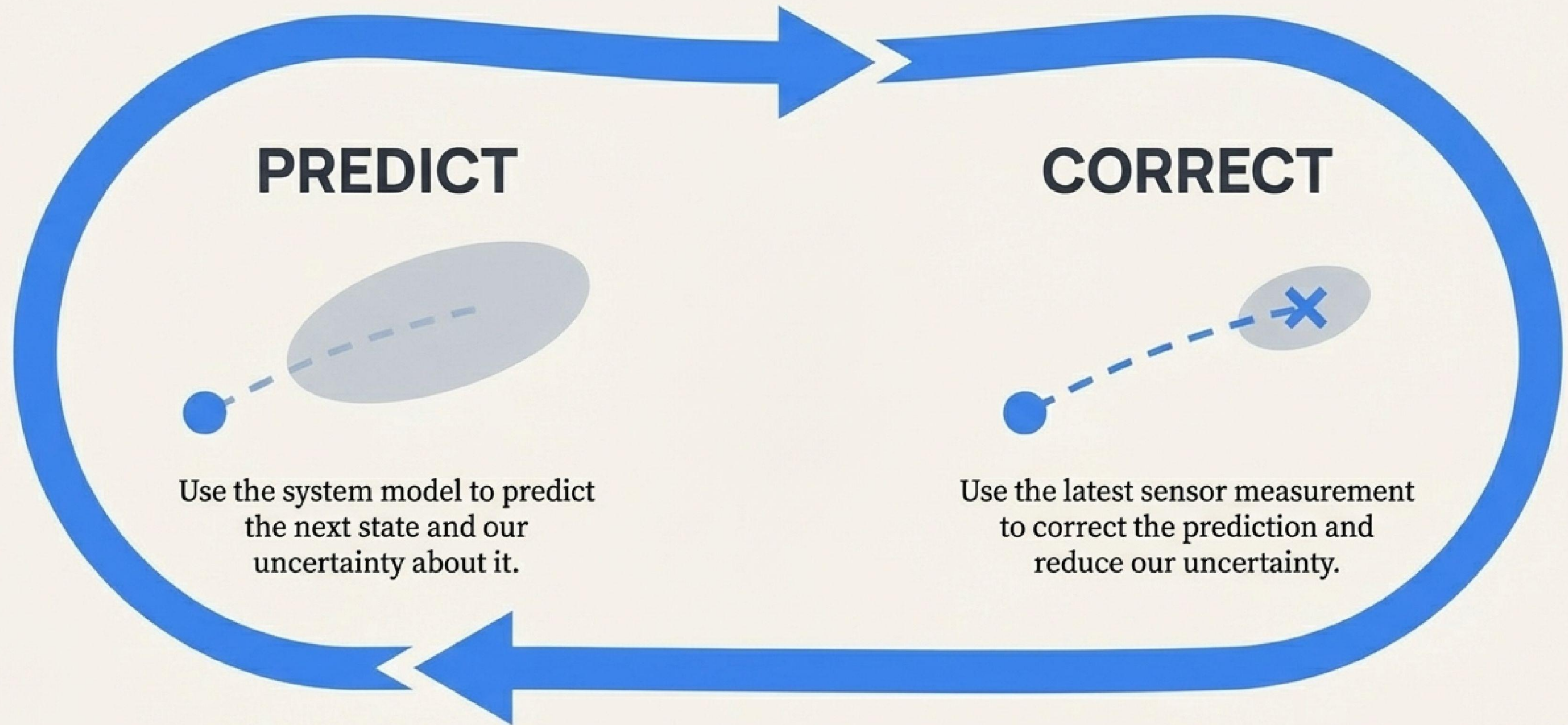


Unobservable States

The true state of a system is often hidden, and we must infer it from the data we can collect.

The Goal: To intelligently fuse our predictions (from the model) with our observations (from sensors) to arrive at the best possible estimate of the true state.

The Core Idea: A Rhythmic Cycle of Prediction and Correction



This two-step process repeats at every time step, continuously refining our estimate.

The Kalman Filter: An Optimal Solution for a Linear World

The Kalman Filter (KF) is the best possible estimator under a specific set of assumptions.

1. Linear System Dynamics

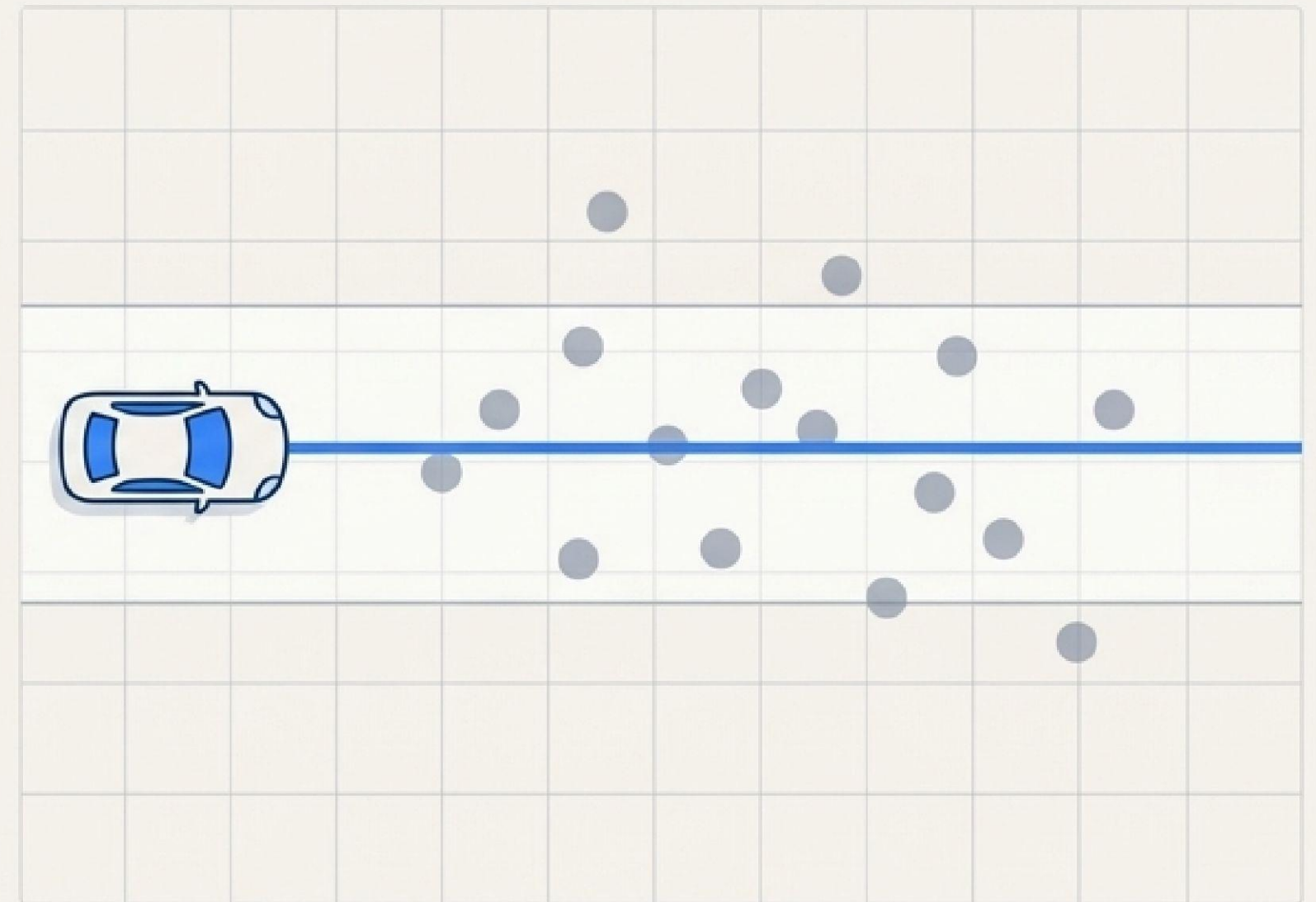
The system's behavior over time can be described with linear equations (e.g., constant velocity).

2. Linear Measurement Model

The relationship between the system's state and the sensor readings is linear.

3. Gaussian Noise

The errors in both the system model and the measurements follow a bell-curve (Gaussian) distribution.



KF estimates the optimal linear path despite noisy sensor data.

The Language of a Linear System

State (Motion) Equation

$$x_k = Ax_{k-1} + Bu_k + w_k$$

The new state.
("Where we think
we are now.")

The state transition
model. ("How the system
evolves on its own.")

The control input
model. ("How we tell
the system to move.")

The process noise.
("Unpredictable
physical disturbances.")

Measurement Equation

$$z_k = Hx_k + v_k$$

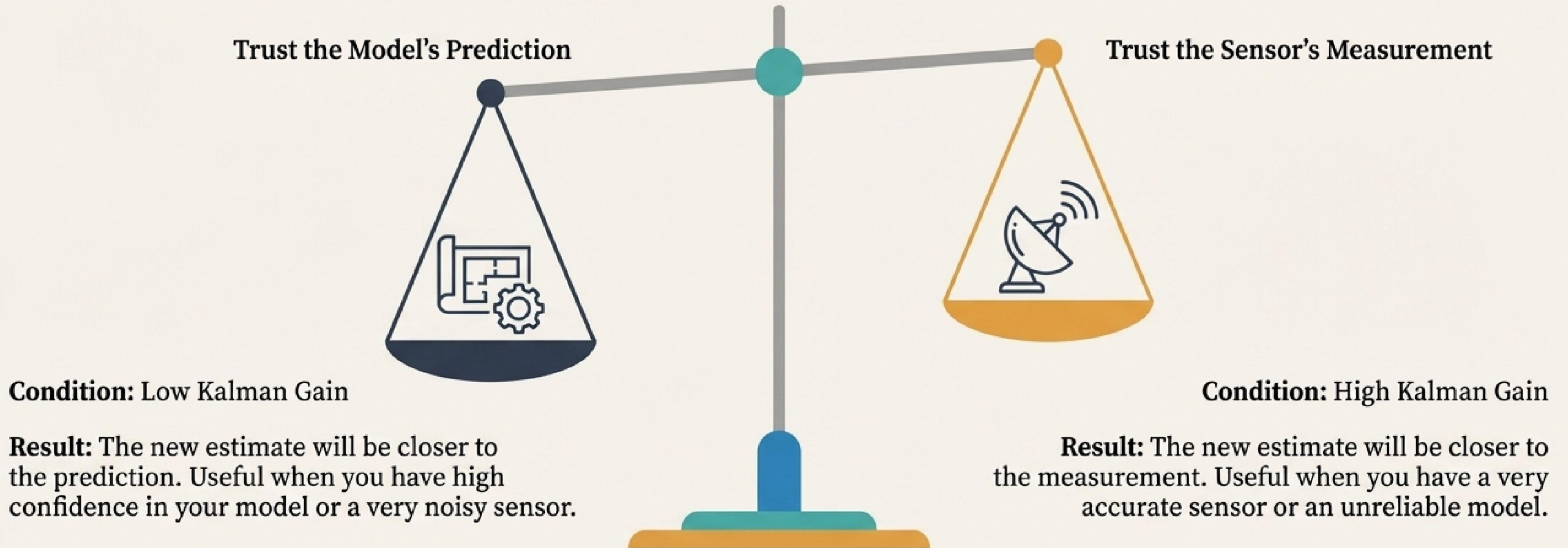
The sensor measurement.
("What our sensor tells us.")

The observation model.
("How the true state relates
to a measurement.")

The measurement noise.
("Inherent sensor error.")

The Deciding Factor: Understanding the Kalman Gain

How much should we trust our prediction versus our new measurement? The Kalman Gain provides the answer.



Key Insight: The Kalman Gain is not static; it is calculated at every step to find the optimal balance.

The Challenge: When the World Isn't a Straight Line

The Limitation

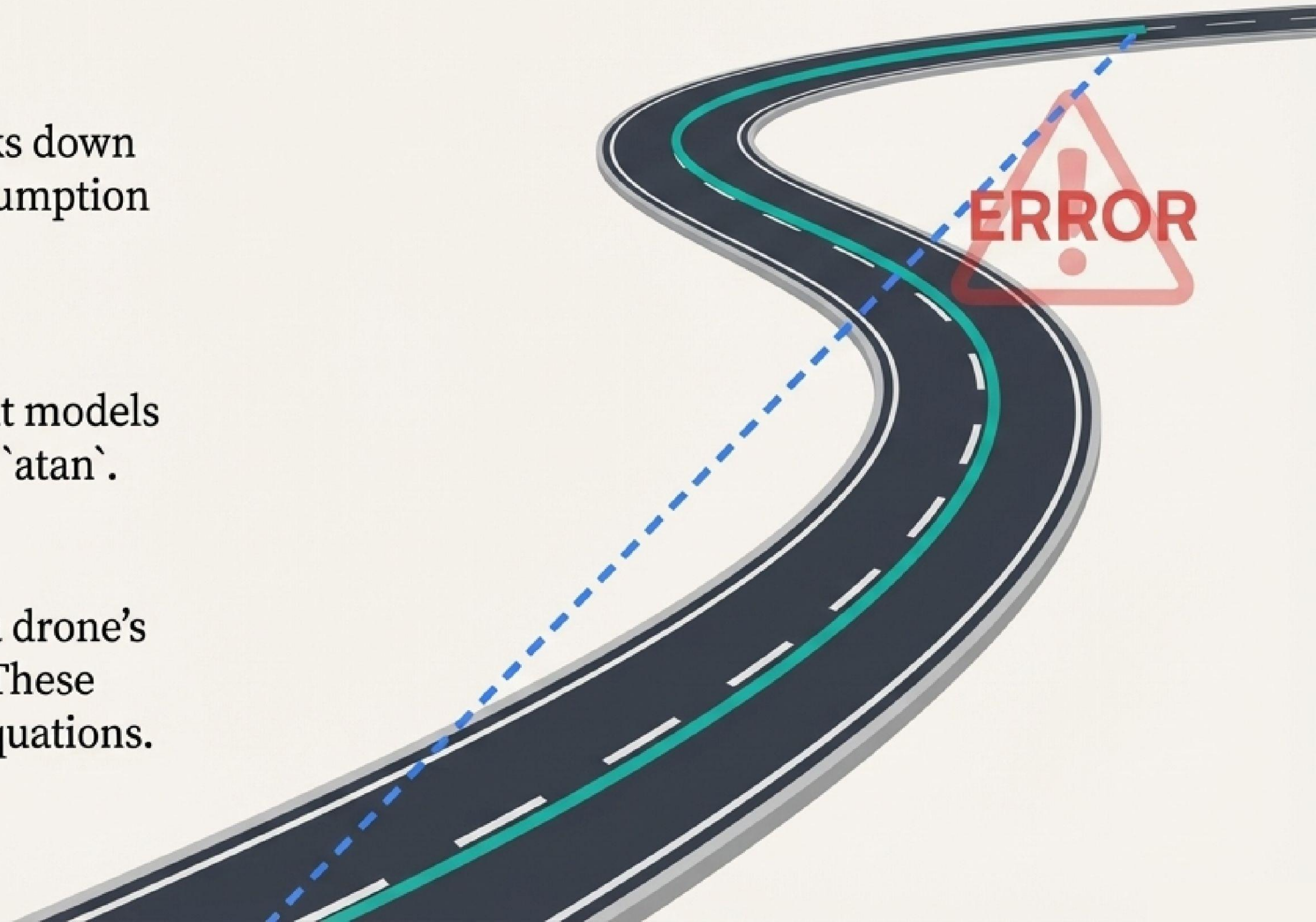
The Kalman Filter's elegant math breaks down when the system is non-linear. The assumption of linearity no longer holds.

What is Non-Linearity?

Systems whose motion or measurement models involve functions like ``sin``, ``cos``, ``√``, or ``atan``.

Practical Example

A car turning, a robot arm rotating, or a drone's altitude based on barometer pressure. These cannot be described by simple linear equations.



The Evolution: The Extended Kalman Filter for Real-World Systems

Core Idea: The EKF adapts the Kalman Filter's framework to handle non-linear models.

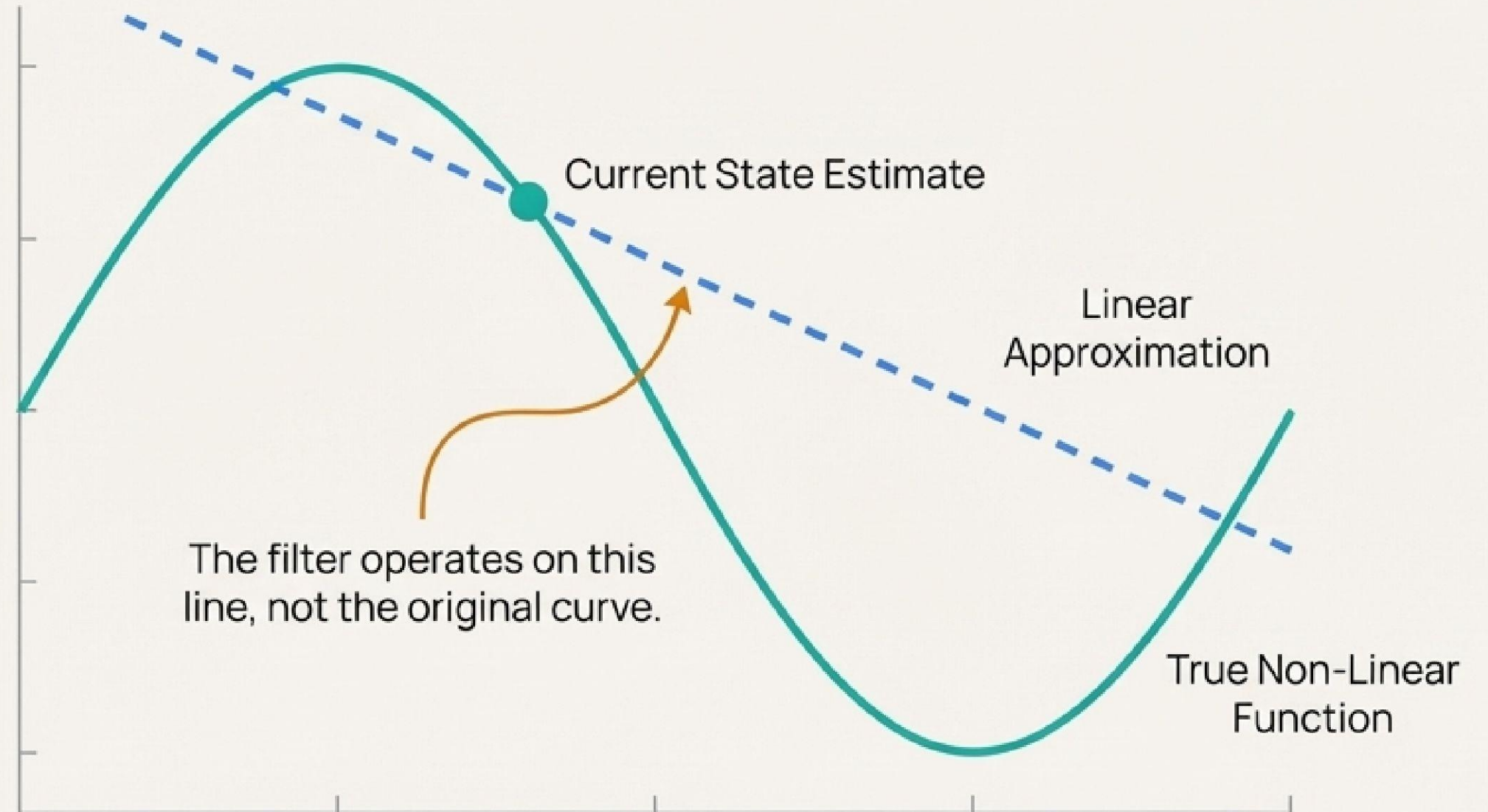
$$\text{EKF} = \text{Kalman Filter} + \text{Linearization}$$

****How it Works**:** The EKF doesn't solve the non-linear problem directly. Instead, at every time step, it approximates the non-linear function with a straight line—a linear approximation—that is valid for the current state estimate. It then applies the standard Kalman Filter logic to this temporary linear model.

The EKF's Trick: Approximating Curves with Lines

The Technique: EKF uses a first-order Taylor expansion to find the best straight-line fit to the non-linear function at the point of the current estimate.

The Tool: This process uses **Jacobian matrices**, which are essentially the multi-dimensional equivalent of a derivative, to find the slope of that best-fit line.



Crucial Caveat: This is only an *approximation* and can diverge if the model is highly non-linear or the initial estimate is poor.

The Language of a Non-Linear System

Key Difference: Instead of the restrictive $Ax + Bu$, we now use general, non-linear functions $f(\cdot)$ and $h(\cdot)$.

Non-Linear Motion Model

$$\mathbf{x}_k = \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k) + \underbrace{\mathbf{w}_k}_{\mathbf{w}_k}$$

$\mathbf{f}(\cdot)$ is a non-linear function that can describe complex dynamics like rotation and translation.

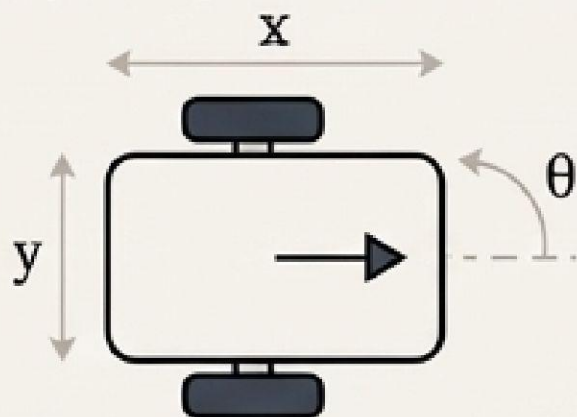
Non-Linear Measurement Model

$$\mathbf{z}_k = \mathbf{h}(\mathbf{x}_k) + \underbrace{\mathbf{v}_k}_{\mathbf{v}_k}$$

$\mathbf{h}(\cdot)$ is a non-linear function that can model sensors like radar, which measures range and bearing.

Example Application: Robot Localization

State: (x, y, θ) - position and orientation



- **Motion:** $\mathbf{f}(\cdot)$ models how wheel encoders update the robot's pose.
- **Measurement:** $\mathbf{h}(\cdot)$ models how a LiDAR sensor reading relates to the robot's position on a map.

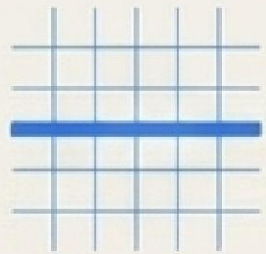
Kalman Filter vs. Extended Kalman Filter: A Head-to-Head Comparison

Feature	Kalman Filter	Extended Kalman Filter
System Type	Strictly Linear	Non-linear
Complexity	Low, computationally efficient	Higher, requires Jacobians
Accuracy	Mathematically optimal (exact)	Approximate, not optimal
Jacobians	Not required	Required for linearization

Core Takeaways for State Estimation



- Both filters are built on the powerful **Predict-Correct** framework to fuse models and measurements.



- The **Kalman Filter** is the simple, stable, and optimal choice for linear systems with Gaussian noise. Think of it as the specialist.



- The **Extended Kalman Filter** adapts this framework for the non-linear systems common in the real world by using linearization. It is the versatile workhorse of robotics, navigation, and autonomous systems.

The choice between them is a trade-off: the mathematical purity of the KF versus the real-world applicability of the EKF.